

South African Labour Market Analysis

New Entrants to Unemployment — QLFS 2026 Q1

Portfolio Analysis Report

Prepared using Python, Pandas, NumPy and Matplotlib

Data source: Statistics South Africa, Quarterly Labour Force Survey (QLFS), 2026 Quarter 1.

1. Executive Summary

This project analyses South Africa's Quarterly Labour Force Survey (QLFS) 2026 Q1 data, focusing on patterns among people with a valid unemployment status who were classified as new entrants to unemployment. The dataset contains **62,819 observations and 200 variables**.

The analysis examined weighted new-entrant shares across provinces, gender groups and education statuses. The strongest provincial contrast was between **Limpopo (51.12%)** and **Western Cape (26.61%)**, a difference of **24.51 percentage points**.

Gender differences also varied by province. Limpopo had the largest female advantage, with a weighted new-entrant share of **55.27% for females** compared with **46.49% for males**, a difference of **8.78 percentage points**.

Education status showed additional variation. **Secondary completed** recorded the highest weighted new-entrant share at **48.37%**, while **Primary completed** recorded the lowest at **35.47%**, a difference of **12.90 percentage points**.

Important interpretation: these results are descriptive associations. They do not establish that province, gender or education causes someone to become a new entrant to unemployment.

2. Research Questions

1. How does the new-entrant share vary across South African provinces?
2. How does the new-entrant share differ between males and females across South African provinces?
3. How does the new-entrant share vary across education statuses?

3. Dataset and Methodology

The project uses the Statistics South Africa QLFS 2026 Q1 unit-record dataset. The working dataset contains 62,819 rows and 200 columns. Analysis was performed in Python.

Tools used

- Pandas — data cleaning, filtering, grouping, aggregation and statistical analysis.
- NumPy — numerical operations and handling missing values.
- Matplotlib — visualisation and chart creation.

Analytical approach

The analysis first inspected variable types and data quality, then investigated survey-specific missing-value codes. New-entrant analysis was restricted to records with a valid **Unempl_Status** value. Survey weights were incorporated when calculating weighted shares.

For a given group, the weighted new-entrant share was calculated as the weighted sum of observations classified as new entrants divided by the total weight of observations with a valid unemployment status in that group.

4. Data Cleaning and Quality Checks

A major cleaning issue was the presence of the extremely large value **1.79769313486232e+308**. This value occurred in variables such as **Hrswrk** and **Unempl_Status** and represented missing or not-applicable observations rather than a real measurement.

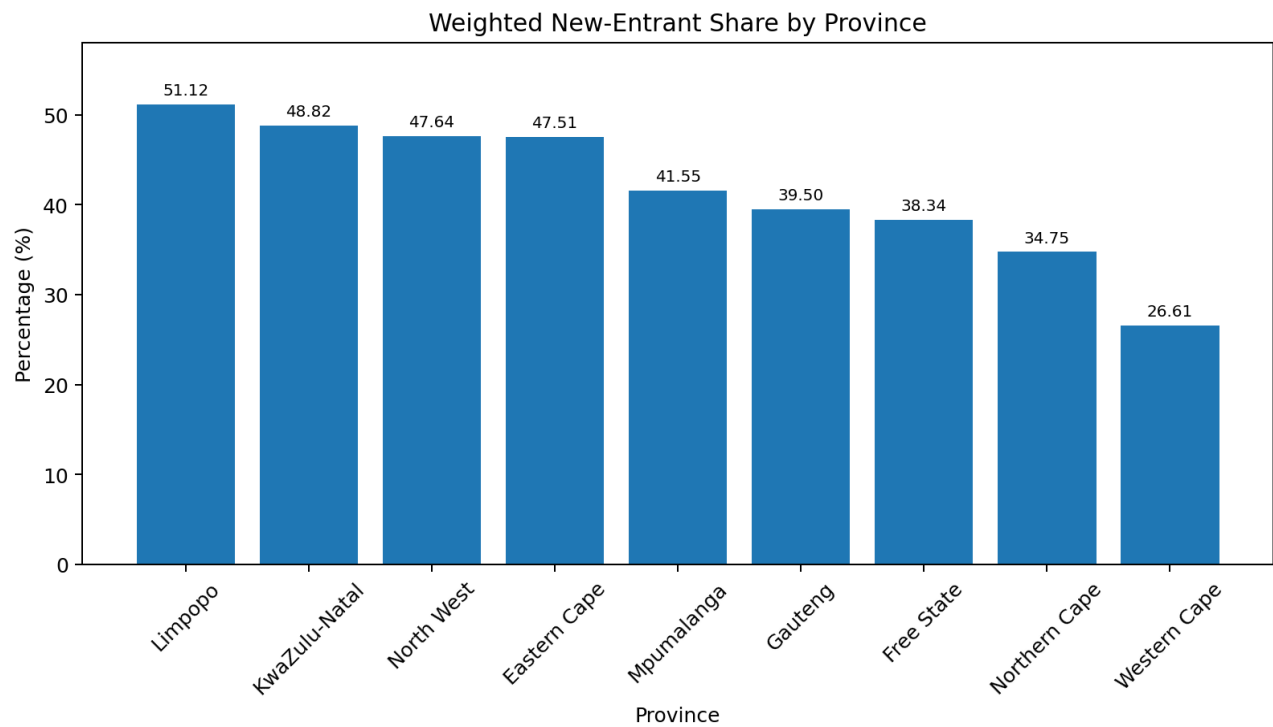
These sentinel values were replaced with **NaN** so that they would not be treated as genuine working-hour or unemployment-status values.

The unemployment-status variable was then filtered to create a valid analytical subset. This was important because using all respondents as the denominator would produce a different quantity and would not represent the share of people with a valid unemployment status who were new entrants.

Other quality checks included inspection of age values, population categories, working-hour values, constant variables and categorical distributions. Rare high ages were not automatically removed because they can be legitimate survey responses.

5. Finding 1 — Provincial Differences

Limpopo recorded the highest weighted new-entrant share at **51.12%**, while Western Cape recorded the lowest at **26.61%**. The difference between the two provinces was **24.51 percentage points**.

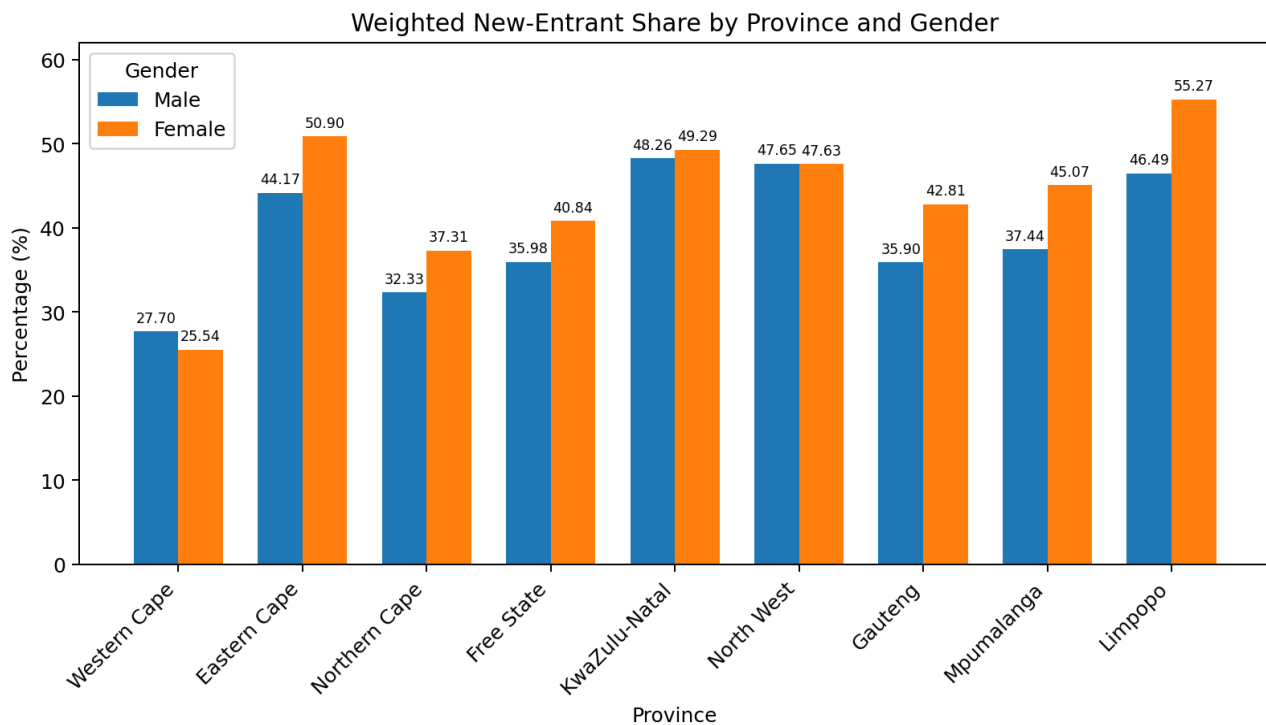


The chart shows substantial variation across provinces. KwaZulu-Natal, North West and Eastern Cape were also above 47%, while Western Cape was notably lower than the other provinces.

Interpretation: provincial context appears to be strongly associated with the composition of unemployment-status records classified as new entrants in this QLFS quarter. The analysis does not, by itself, identify the causes of these differences.

6. Finding 2 — Gender Differences by Province

Gender differences were not uniform across provinces. Limpopo showed the largest female advantage: **55.27% for females** versus **46.49% for males**, a difference of **8.78 percentage points**.

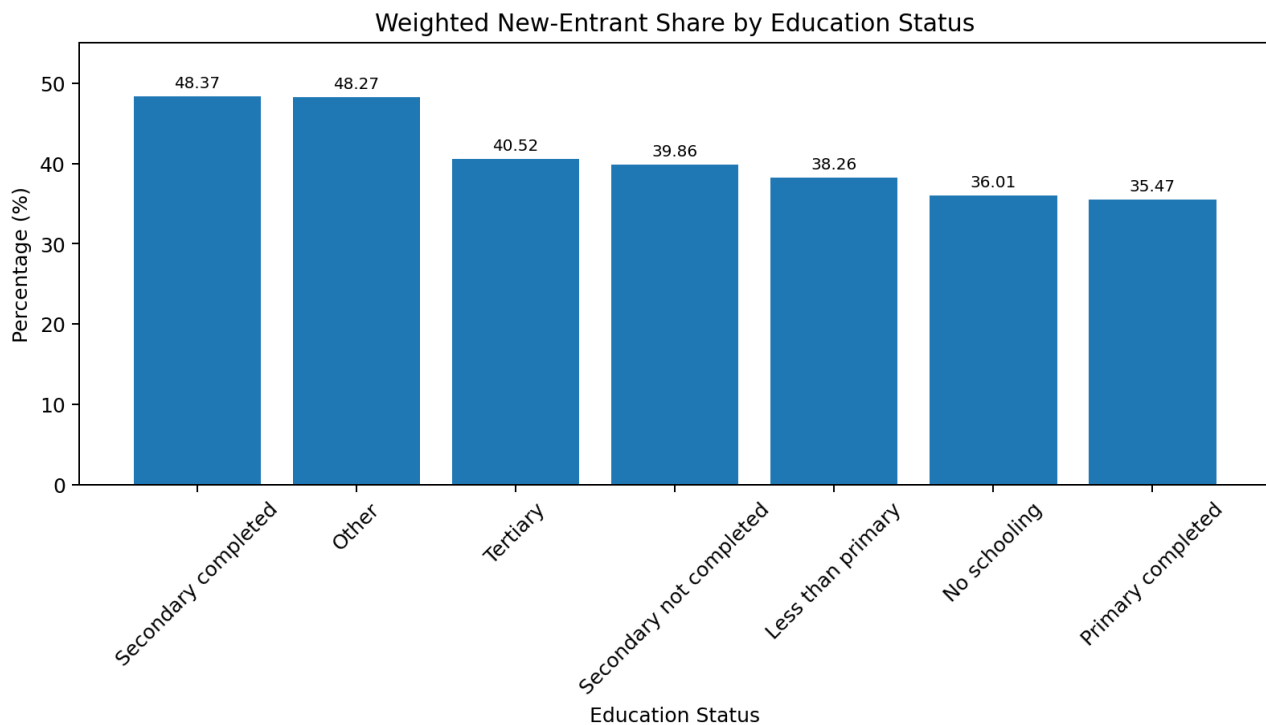


Province	Male (%)	Female (%)	Female – Male (pp)
Western Cape	27.70	25.54	-2.16
Eastern Cape	44.17	50.90	6.73
Northern Cape	32.33	37.31	4.98
Free State	35.98	40.84	4.86
KwaZulu-Natal	48.26	49.29	1.03
North West	47.65	47.63	-0.02
Gauteng	35.90	42.81	6.91
Mpumalanga	37.44	45.07	7.63
Limpopo	46.49	55.27	8.78

Western Cape was the main exception in the opposite direction, with females at 25.54% compared with 27.70% for males. North West showed almost no difference between genders.

7. Finding 3 — Education Status

Secondary completed had the highest weighted new-entrant share at **48.37%**, closely followed by Other at 48.27%. Primary completed had the lowest share at **35.47%**. The difference between the highest and lowest categories was **12.90 percentage points**.



The results suggest variation across education categories, but the relationship is not a simple monotonic pattern. In particular, the Other category is relatively small and should be interpreted with care.

8. Limitations

- 1. Descriptive analysis:** The results identify patterns and associations rather than causal effects.
- 2. Survey weighting:** Weighted percentages depend on the survey weights supplied with the QLFS. They should not be interpreted as simple sample percentages.
- 3. Denominator:** The new-entrant percentages reported in this project are shares among records with a valid unemployment status. They are not presented as the overall South African unemployment rate.
- 4. Missing and not-applicable values:** Survey datasets can encode missingness using special values. Incorrectly treating such codes as real measurements can substantially distort results.
- 5. Single quarter:** This analysis uses QLFS 2026 Q1. It describes one quarter and cannot by itself establish a long-term trend.
- 6. Small categories:** Some education categories, particularly Other, contain fewer observations than the largest categories and may require additional checking before drawing strong conclusions.

9. Conclusion

The QLFS 2026 Q1 analysis found meaningful differences in the weighted share of unemployment-status records classified as new entrants across province, gender and education groups.

The largest provincial contrast was between Limpopo (51.12%) and Western Cape (26.61%). Limpopo also showed the largest female advantage, with females at 55.27% compared with 46.49% for males. Across education categories, Secondary completed recorded the highest share at 48.37%, compared with 35.47% for Primary completed.

Overall, the project demonstrates a complete data-analysis workflow: understanding a real-world survey dataset, identifying survey-specific missing values, cleaning the data, applying an appropriate analytical denominator, incorporating weights, answering research questions and communicating results visually.

10. Portfolio Presentation Notes

For a public portfolio, the analysis is strongest when accompanied by the cleaned notebook, a README, the three final charts and a short explanation of the methodology. The notebook should retain the code used to create the reported figures so that the results are reproducible.

Recommended project structure:

QLFS-2026-Data-Analysis/

■■■■ README.md

■■■■ data/

■■■■ notebooks/QLFS_2026_Analysis.ipynb

■■■■ visuals/

■■■■ report/QLFS_2026_Q1_New_Entrants_Report.pdf

■■■■ requirements.txt

Official data source: Statistics South Africa — Quarterly Labour Force Survey (QLFS), 2026 Q1. The QLFS portal provides access to unit-record data, metadata, questionnaires and related documentation.